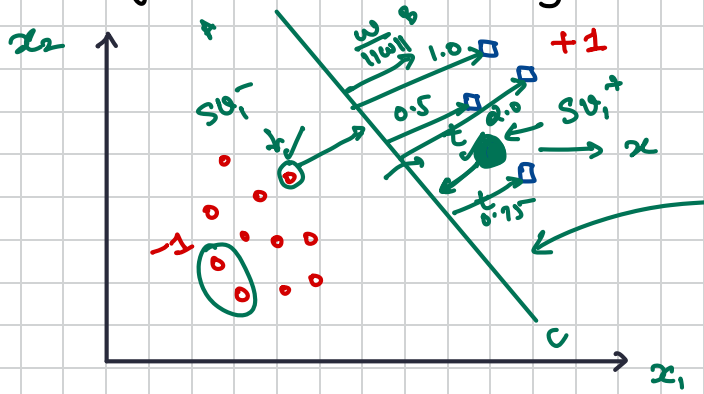


Support vector Machines

→ High-dimensional linearly non-separable problems in $\mathbb{R}^p$



$\frac{t \times w}{\|w\|}$

w, b ??

$y = mx + c$

$mx - y + c = 0$

$\begin{bmatrix} m \\ -1 \end{bmatrix}^T \begin{bmatrix} x \\ y \end{bmatrix} + b = 0$

$w^T x + b = 0$

$x = y$

$x - y = 0$

$\begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$

$\begin{bmatrix} 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & -1 \end{bmatrix}$

$= 0$

$w^T x + b = 0$

$w^T x + b > 0 \quad y = +1$

$w^T x + b < 0 \quad y = -1$

$w^T x^{sv} + b > 0$

$y (w^T x + b) > 0$

$\left(w^T x - t \frac{w}{\|w\|} \right) + b = 0$

$w^T x + b = \frac{t\, w}{\|w\|}$

$t =$

$t = \|w\| (w^T x + b) w^{-1}$

$t = \frac{w x + b}{\|w\|}$

dis. of any point from line

① → $w^T x + b \equiv y = mx + c$

max t for point that is closest

$t(w, b)$

$\max_{w,b} \left(\min_{y_i} t_i \right)$

w, b

(2, 1, 0.5, 4.5)

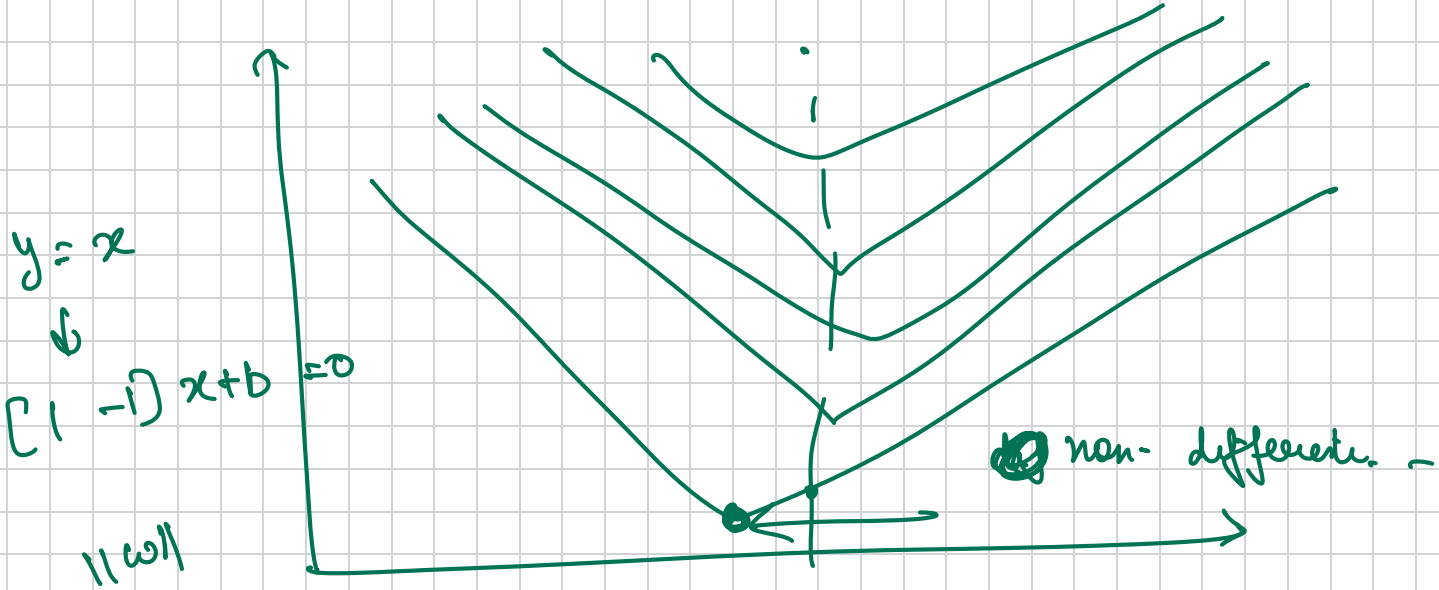
↓

(0.5)

$\max_{w,b}$

w^*, b^*

∀ all w, b:



$w^T x + b = 0$

$y = +1$

$y = -1$

$(w^T x + b)(y) \geq 0$

$$\max \left\{ \min_i \frac{(w^T x_i + b)(y_i)}{\|w\|} \right\}$$

$(w^T x + b)$

$w = 2w$

$b = 2b$

– Find those values of w, b that makes

$(w^T x + b)(y) = 1$ for nearest y

$(w^T x + b)(y) > 1$ for all other ys

$\rightarrow (w^T x + b)(y) \geq 1$

$$\max \frac{1}{\|w\|} \equiv \min \|w\| = \frac{1}{2} \min \|w\|^2$$

s.t.

$\min_i (w^T x + b) y \geq 1$ $\forall$ all i

$(w^T x + b)(y) - 1$

$$J(w, b, \alpha_i) \equiv \min \left\{ \frac{1}{2}\|w\|^2 - \sum_{i=1}^{n} \alpha_i \left((w^T x + b) y - 1 \right) \right\}$$

$\alpha \geq 0$

$\frac{1}{2}\|w\|^2$ ↑ weight

$\sum_{i=1}^{n}$ ↓ M

α_i ↑ Cons.

Find those αs which are supports?

Game

$$\left\{ \max_{\alpha} \left\{ \min_{w,b} \right\} \right\} = \left\{ \max \min \left\{ \min_{w,b} \left\{ \max_{\alpha} \right\} \right\} \right\}$$

1 violating

$$(w^T x + b)(y) - 1 < 0 \quad \checkmark$$

Strictly convex & there is a feasible sol^n

One Boundary, Two Scalings, the Same Margin

Problem. Eight road segments have two features: rainfall index x_1 and wetness index x_2. Their labels are $y = +1$ (flooded) and $y = -1$ (passable). Find a maximum-margin separating line. This example assumes **linearly separable data**.

Flooded ($y = +1$)	$(3,3)$	$(4,3)$	$(3,4)$	$(4,4)$
Passable ($y = -1$)	$(1,1)$	$(0,1)$	$(1,0)$	$(0,0)$

1. Try a separating line and calculate its scores

Choose $x_1 + x_2 = 4$. In the notation $w^\top x + b = 0$, this means $w = (1,1)^\top$, $b = -4$, and $\|w\| = \sqrt{2}$.

$$s(x) = w^\top x + b = x_1 + x_2 - 4, \qquad d_{\text{signed}}(x) = \frac{s(x)}{\|w\|}.$$

A positive score predicts flooded; a negative score predicts passable. Ordinary distance is $|d_{\text{signed}}|$.

2. Identify the nearest points, not the largest scores

Point x	Label y	Score s	ys	Ordinary distance
$(3,3)$	$+1$	2	2	$2/\sqrt{2} = \sqrt{2}$
$(1,1)$	-1	-2	2	$2/\sqrt{2} = \sqrt{2}$
$(4,3)$	$+1$	3	3	$3/\sqrt{2}$
$(0,1)$	-1	-3	3	$3/\sqrt{2}$

The remaining points have label-adjusted scores $3, 4, 3, 4$. Thus the smallest $y_i s(x_i)$ is 2, attained at $(3,3)$ and $(1,1)$. The **one-sided geometric margin** is $\gamma = \min_i y_i s(x_i)/\|w\| = 2/\sqrt{2} = \sqrt{2}$.

3. Make the nearest label-adjusted score equal to 1

Divide *both* w and b by 2: $\widetilde{w} = (0.5, 0.5)^\top$ and $\widetilde{b} = -2$.

$$0.5x_1 + 0.5x_2 - 2 = 0 \iff x_1 + x_2 = 4, \qquad \|\widetilde{w}\| = 1/\sqrt{2}.$$

The nearest scores become $+1$ and -1, so their label-adjusted scores are both 1. Every other label-adjusted score is 1.5 or 2. Hence $y_i(\widetilde{w}^\top x_i + \widetilde{b}) \geq 1$ for every point. **The line has not moved:** the new margin is $1/\|\widetilde{w}\| = \sqrt{2}$, exactly as before.

4. Connect these numbers to the SVM objective

Once this scaling is fixed, maximizing the margin is equivalent to

$$\boxed{\min_{w,b} \frac{1}{2}\|w\|^2 \quad \text{subject to} \quad y_i(w^\top x_i + b) \geq 1 \quad \text{for every } i.}$$

For our candidate, the objective is $\frac{1}{2}(0.5^2 + 0.5^2) = \boxed{0.25}$. Equivalent objectives have the same optimizer; their numerical values need not be equal.

5. Why is this line actually optimal?

The opposite-class points $(3,3)$ and $(1,1)$ are $2\sqrt{2}$ apart. A line with margin γ must place each at least γ away on opposite sides. Their perpendicular separation is therefore at least 2γ, but cannot exceed their distance $2\sqrt{2}$. Thus $\gamma \leq \sqrt{2}$. Our line attains this bound, so it is maximum-margin. These two points are its **support vectors**. The full corridor between the margin lines is $2\gamma = 2\sqrt{2}$ wide.

Takeaway: Rescaling changes scores and coefficient values, not the boundary or its distances. The constraint fixes that arbitrary scale; minimizing the norm then selects the widest margin.